

AI-ENABLED STUDENT ADMISSION SYSTEM

A Salesforce-Based Intelligent Admission Management Solution

Project Report

Submitted as part of the Salesforce Developer Program

Prepared by

Annamneedi Satyanarayana
siddunaidu592@gmail.com

Platform: Salesforce Developer Edition

Technologies: Flow Builder · Agentforce · Prompt Builder · Reports & Dashboards

Table of Contents

User Story.....	3
1. Introduction.....	4
1.1 Project Overview.....	4
1.2 Purpose.....	4
1.3 Scope of the Document.....	4
2. Ideation Phase.....	5
2.1 Problem Statement.....	5
2.2 Stakeholder Mapping.....	5
2.3 Brainstorming.....	6
3. Requirement Analysis Phase.....	7
3.1 Requirement Gathering & Planning.....	7
3.2 Functional Scope.....	7
3.3 Non-Functional Requirements.....	8
3.4 Data Flow.....	9
3.5 Technology Stack.....	9
4. Project Design Phase.....	10
4.1 Problem–Solution Fit.....	10
4.2 Proposed Solution.....	10
4.3 Solution Architecture.....	10
5. Project Planning & Scheduling.....	11
5.1 Execution Roadmap.....	11
5.2 Product Backlog & Sprint Plan.....	11
6. Project Development Phase (Milestones 1–8).....	13
7. Agentforce AI Implementation.....	23
8. Reports & Dashboards.....	26
9. Security & Access Control.....	28
10. Functional & Performance Testing.....	32
11. Maintenance, Monitoring & Troubleshooting.....	32
12. Advantages & Disadvantages.....	33
13. Conclusion.....	33
14. Future Scope.....	33
15. Appendix.....	34

User Story

As an applicant, I want to submit my admission application and track its status so that I know the progress of my admission process.

As an admission officer, I want the system to automatically identify incomplete applications and missing documents so that I can review applications efficiently.

I want an AI assistant that can analyze application details and generate admission guidance so that I can process applications faster while keeping a human reviewer in control.

As an admission manager, I want dashboards showing application status, program demand, and review performance so that I can monitor the admission process.

As a system administrator, I want role-based access control so that applicant and admission information remains secure.

1. Introduction

1.1 Project Overview

The “AI-Enabled Student Admission System” is a Salesforce-based admission management solution built to help academic institutions manage, evaluate, and process student admission applications more efficiently. The system is implemented entirely using core Salesforce Administration, Flow Builder, and Agentforce capabilities within a Salesforce Developer Org, without relying on external AI services or custom Apex code.

At its core, the application centralizes admission operations around a custom Admission Application object, extended with custom objects for Program, Document Verification, and AI Feedback. Record-triggered flows automatically evaluate application completeness, check program eligibility, verify required documents, and assign application status the moment an application is created or updated. An Agentforce-powered assistant, built using Prompt Builder and a custom Agent Topic, reads the application details and generates admission guidance that an admission officer can review and act upon — significantly reducing processing time while keeping a human reviewer in the loop.

Role-based security, implemented through cloned profiles and dedicated permission sets, ensures that admission officers, admission managers, and administrators each have access appropriate to their responsibilities. Reports and dashboards give admission managers real-time visibility into application volume, program-wise demand, and review performance, while an AI feedback loop captures whether admission officers found the AI's guidance useful — creating a foundation for continuous improvement.

1.2 Purpose

The purpose of this project is to:

- Centralize and standardize how student admission applications are logged, tracked, and processed.
- Automate repetitive administrative work — completeness verification, eligibility checks, and status assignment — that would otherwise be done manually by admission officers.
- Assist admission officers with AI-generated admission guidance so reviews are faster, more consistent, and more informative.
- Provide admission managers with real-time, data-driven visibility into application volume and review performance.
- Protect applicant and admission data through role-based access control, field-level security, and validation rules.
- Establish a feedback mechanism that measures whether AI-generated guidance is genuinely useful to admission officers, enabling continuous refinement of the system.

1.3 Scope of the Document

This document covers the complete lifecycle of the project — from requirement gathering and stakeholder mapping, through system design, data modeling, automation, AI integration, and security configuration, to testing, reporting, and future scope. Screenshots of the actual implementation in the Salesforce Developer Org, along with clearly labeled placeholders for screenshots to be captured, are included throughout to document the functionality that has been designed and built.

2. Ideation Phase

2.1 Problem Statement

Academic institutions that rely on manual processes for admission handling face recurring, predictable pain points. The table below captures these pain points in a structured problem-statement format, identifying who is affected, what they are trying to do, what is blocking them, why the blocker exists, and how it makes them feel.

PS	I am (Applicant/User)	I'm trying to	But	Because	Which makes me feel
PS-1	Admission Officer	Review student applications quickly and correctly	Applications are not evaluated or prioritized consistently	Completeness and eligibility checks are done manually	Overwhelmed and inefficient
PS-2	Admission Manager	Monitor application volume and review performance	Reports are delayed and scattered across records	There is no centralized reporting on admission data	Uncertain and reactive
PS-3	Applicant	Get a fast, accurate update on my admission status	Status updates are slow and inconsistent	Officers evaluate every application manually from scratch	Anxious and undervalued

2.2 Stakeholder Mapping

Stakeholder mapping identifies the key individuals and groups involved in the development, implementation, and day-to-day use of the system. Each stakeholder group has distinct responsibilities and interests in the project's success.

Stakeholder Group	Role in the System
Applicants / Students	Submit admission applications, upload documents, and receive status updates and AI-assisted guidance.
Admission Officers	Review applications, verify documents, act on AI-generated admission guidance, and record feedback on guidance quality.
Admission Managers	Monitor application volume, program demand, and officer review performance using reports and dashboards.
System Administrators	Configure objects, automation, security, and Agentforce settings; maintain the system.
Institution Stakeholders	Define requirements and validate that the solution aligns with institutional admission goals.

2.3 Brainstorming

Before settling on the final approach, several alternative solutions were considered for automating student admission application handling:

- Continue with a fully manual, spreadsheet-based application log.
- Build a simple email-based application intake workflow with no eligibility logic.
- Use a third-party admission management tool with a paid AI add-on.
- Build a Salesforce-native admission system using standard objects and Flow automation only, without AI.
- Build a Salesforce-native admission system enhanced with Agentforce and Prompt Builder for AI-assisted completeness evaluation and admission guidance (selected idea).

Selected Idea: A Salesforce-based AI-Enabled Student Admission System, built on core Admin and Flow Builder features and enhanced with Agentforce for AI-assisted application handling — chosen for its low implementation cost, strong scalability, native security model, and ability to demonstrate real-world Salesforce and Agentforce skills.

3. Requirement Analysis Phase

3.1 Requirement Gathering & Planning

The requirement gathering phase focused on understanding the challenges faced by academic institutions in processing student admission applications efficiently. Stakeholders required a centralized system to manage applicant information, reduce manual effort in eligibility and completeness checks, ensure timely status updates, and improve admission officer productivity. The system needed to automatically verify application completeness, check program eligibility, assist officers with AI-generated guidance, and provide visibility into admission performance through reports.

Based on these discussions, functional and non-functional requirements were identified, emphasizing ease of use, automation, data accuracy, role-based access control, and scalability. The solution was designed to be implemented using core Salesforce features in a Developer Org, along with Agentforce and Prompt Builder for AI capabilities, ensuring simplicity, reliability, and ease of maintenance.

3.2 Functional Scope

The functional scope of the project defines the key features and capabilities delivered by the Salesforce-based student admission system. The application allows users to create and manage admission applications, automatically verify completeness, check program eligibility, and identify missing documents.

Admission officers can view AI-generated admission guidance to review applications, update application status, and communicate with applicants. The system also captures officer feedback on the AI's guidance to evaluate system effectiveness. Role-based access control ensures that admission officers, admission managers, and administrators have appropriate permissions. Managers can monitor admission performance using reports, while administrators can configure automation, security, and the data model.

Functional Requirements

FR No.	Requirement
FR-1	Admission application creation and management with automatic completeness verification
FR-2	Automatic program eligibility check based on academic percentage
FR-3	Automatic identification of missing documents
FR-4	AI-generated admission guidance for each application using Agentforce and Prompt Builder
FR-5	Admission officer feedback capture on AI guidance quality
FR-6	Role-based access control via profiles and permission sets
FR-7	Reports and dashboards for application status and review performance monitoring
FR-8	Automated email notifications when admission status changes

3.3 Non-Functional Requirements

NFR-1: Usability

The system should be easy to use for applicants, admission officers, managers, and administrators without requiring technical training.

- Simple, intuitive Lightning interface with clear navigation.
- Officer-facing AI guidance surfaced directly on the Admission Application record.
- A new admission officer should be able to open an application, view the AI Admission Guidance, and act within minutes.

NFR-2: Security

The system must protect applicant and admission data from unauthorized access.

- Role-based access control through cloned profiles (Admission Officer, Admission Manager).
- Field-level security restricting edit access to sensitive fields such as Admission Decision and AI Admission Guidance.
- Dedicated permission sets that gate access to AI features and reporting tools.

NFR-3: Reliability

The system should work correctly at all times and store data accurately without loss.

- Record-triggered flows execute reliably on application creation and update.
- Validation rules prevent incomplete or inconsistent data from being saved.
- Salesforce's cloud infrastructure ensures automatic backups and data durability.

NFR-4: Performance

The system should respond quickly and handle application volumes efficiently.

- Completeness verification and eligibility evaluation complete instantly on save via record-triggered flows.
- Reports and dashboards render within a few seconds even with a growing application volume.

NFR-5: Availability

The system should be accessible whenever admission officers need it.

- Hosted entirely on Salesforce's cloud platform with high uptime guarantees.
- Accessible from desktop and mobile Salesforce clients.

NFR-6: Scalability

The system should support growth in application volume and admission team size without redesign.

- Custom objects and flows are built generically so new programs, document types, or officers can be added without structural changes.
- Salesforce's multi-tenant architecture scales automatically as data volume grows.

3.4 Data Flow

The system follows a simple, linear data flow: an applicant or admission officer creates an Admission Application, which is evaluated by validation rules and record-triggered flows for completeness, program eligibility, and document verification. The Agentforce agent then reads the application and produces admission guidance via a Prompt Template. Once reviewed, an email alert notifies the applicant of the status change, and the application data feeds into reports and dashboards for management visibility.

Application Creation → Validation Rules → Record-Triggered Flow (Completeness, Eligibility, Document Check) → Agentforce / Prompt Builder (Admission Guidance) → Officer Review & Decision → Email Alert → Reports & Dashboards.

3.5 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning Experience
Database	Salesforce Cloud (Standard & Custom Objects)
Automation	Record-Triggered Flows, Validation Rules, Email Alerts
AI / Intelligence	Agentforce, Prompt Builder, Prompt Templates
Security	Profiles, Permission Sets, Field-Level Security
Analytics	Reports & Dashboards

4. Project Design Phase

4.1 Problem–Solution Fit

Manual admission processing is prone to inconsistent evaluation, missed documents, and slow, inconsistent applicant communication. The proposed solution replaces this with automated completeness and eligibility checks through record-triggered flows, and augments officer reviews with AI-generated admission guidance via Agentforce. This ensures every application is evaluated consistently, missing documents are identified promptly, and applicants receive timely status communication.

4.2 Proposed Solution

Capability	Description
Application Management	Create, view, update, and track student admission applications end-to-end.
Automatic Completeness Check	Record-triggered flow evaluates application details and determines completeness.
Program Eligibility Check	Flow automatically compares academic percentage against program eligibility criteria.
AI Admission Guidance	Agentforce Action backed by a Prompt Template drafts clear, professional admission guidance.
AI Feedback Capture	Officers log whether the AI guidance was helpful, feeding a continuous-improvement loop.
Validation Rules	Enforce mandatory fields, such as Academic Percentage before submission.
Email Alerts	Automated status-update email sent to the applicant when application status changes.
Role-Based Security	Profiles and permission sets control access for officers, managers, and administrators.
Reports & Dashboards	Real-time visibility into applications by status, program, and officer review performance.

4.3 Solution Architecture

The solution follows a layered architecture that keeps the presentation, automation, data, AI, and analytics concerns cleanly separated, which makes the system easier to maintain and extend:

- Presentation Layer — Salesforce Lightning App and Admission Application UI
- Application Layer — Record-Triggered Flows and Validation Rules
- AI Layer — Agentforce Agent, Topic, Action, and Prompt Template
- Data Layer — Standard and Custom Objects (Contact, Admission Application, Program, Document Verification, AI Feedback)
- Security Layer — Profiles, Permission Sets, Field-Level Security
- Analytics Layer — Reports and Dashboards

This layered architecture ensures modularity, scalability, and maintainability — each layer can be modified independently without disrupting the others.

5. Project Planning & Scheduling

5.1 Execution Roadmap

The project was executed in eight structured phases to ensure smooth development, testing, and deployment:

Phase	Focus Area
Phase 1	Requirement Analysis & Planning — understand institutional needs, map stakeholders, finalize scope and data model.
Phase 2	Environment Setup — create the Salesforce Developer Org and enable Lightning Experience.
Phase 3	Data Model & Security — create objects, fields, relationships, profiles, and permission sets.
Phase 4	Automation & Evaluation — build record-triggered flows for completeness, eligibility, and document checks.
Phase 5	Officer Assistance & Feedback — configure Agentforce, Prompt Builder, and the AI feedback loop.
Phase 6	Reporting & Monitoring — build reports for applications by status, program, and officer review performance.
Phase 7	Testing & Validation — verify automation, data integrity, and AI guidance across scenarios.
Phase 8	Deployment & Documentation — activate configurations and finalize documentation for demonstration.

5.2 Product Backlog & Sprint Plan

The project followed an Agile-based approach, broken into sprints aligned with the milestones of the execution roadmap.

Sprint	Epic	User Story	Points	Priority
Sprint-1	Environment Setup	As an admin, I want to create a Salesforce Developer Org so that I can configure and deploy the admission application.	1	High
Sprint-1	Data Modeling	As an admin, I want to create Admission-related standard and custom objects so admission data can be stored systematically.	3	High
Sprint-2	Data Modeling	As a user, I want dedicated tabs and a Lightning App so I can navigate admission records easily.	3	High
Sprint-2	Data Modeling	As an admin, I want fields and relationships across Admission Application, Program, Document Verification, and AI Feedback so complete admission data is captured.	5	High

Sprint	Epic	User Story	Points	Priority
Sprint-3	Automation	As a system, I want to validate application data so an application cannot be submitted without the applicant's academic percentage.	3	High
Sprint-3	Automation	As a system, I want a record-triggered flow to check completeness, evaluate eligibility, and identify missing documents automatically.	5	High
Sprint-4	AI Integration	As an officer, I want an Agentforce assistant to generate admission guidance for an application so I can review it faster.	5	High
Sprint-4	AI Integration	As an officer, I want to log feedback on AI guidance so the system's effectiveness can be tracked.	3	Medium
Sprint-5	Automation	As a system, I want to send an automated email when admission status changes so applicants are notified of progress.	3	High
Sprint-5	Security	As an admin, I want profiles and permission sets so officers, managers, and admins have appropriate access.	5	High
Sprint-6	Reporting	As a manager, I want reports on applications by status and program so I can monitor admission demand.	4	High

6. Project Development Phase

The development phase was executed across eight milestones, moving from environment setup through data modeling, automation, AI integration, and email notifications. Each milestone below is documented with the configuration steps performed and a screenshot or clearly labeled placeholder verifying the intended implementation.

Milestone 1 – Salesforce Developer Account & Environment Setup

A Salesforce Developer Org was created to serve as the environment for building and testing the entire application, free of cost and fully featured for configuration, automation, and Agentforce.

- Signed up at developer.salesforce.com/signup with role, institution, and country details.
- Verified the account via the activation email and set a secure password and security question.
- Landed on the Salesforce Setup Home page, confirming the org was ready for configuration.

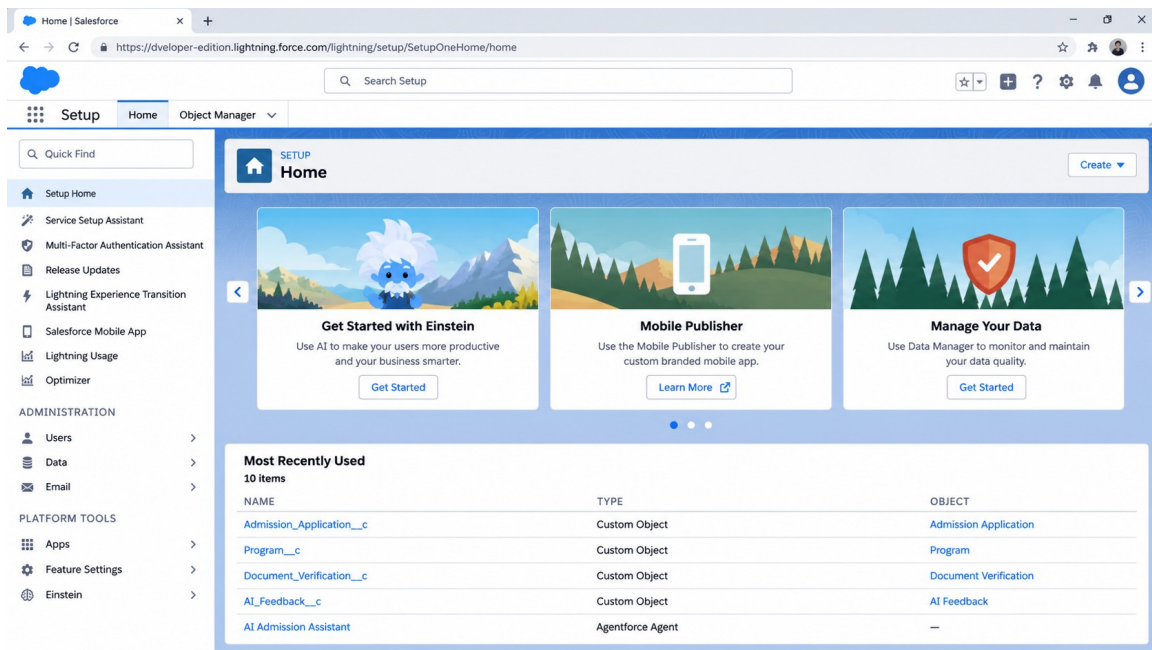


Figure 1: Salesforce Setup Home page after successful Developer Org activation.

Milestone 2 – The Lightning App

A dedicated Lightning App, “AI-Enabled Student Admission System,” was created to bring together all the objects, reports, dashboards, and the Agentforce assistant into a single, officer-friendly workspace.

- Configured app name, branding, and description in App Manager.
- Added Admission Application, Program, Document Verification, AI Feedback, Reports, and Dashboards as navigation items.
- Assigned the System Administrator profile for app access and saved the configuration.

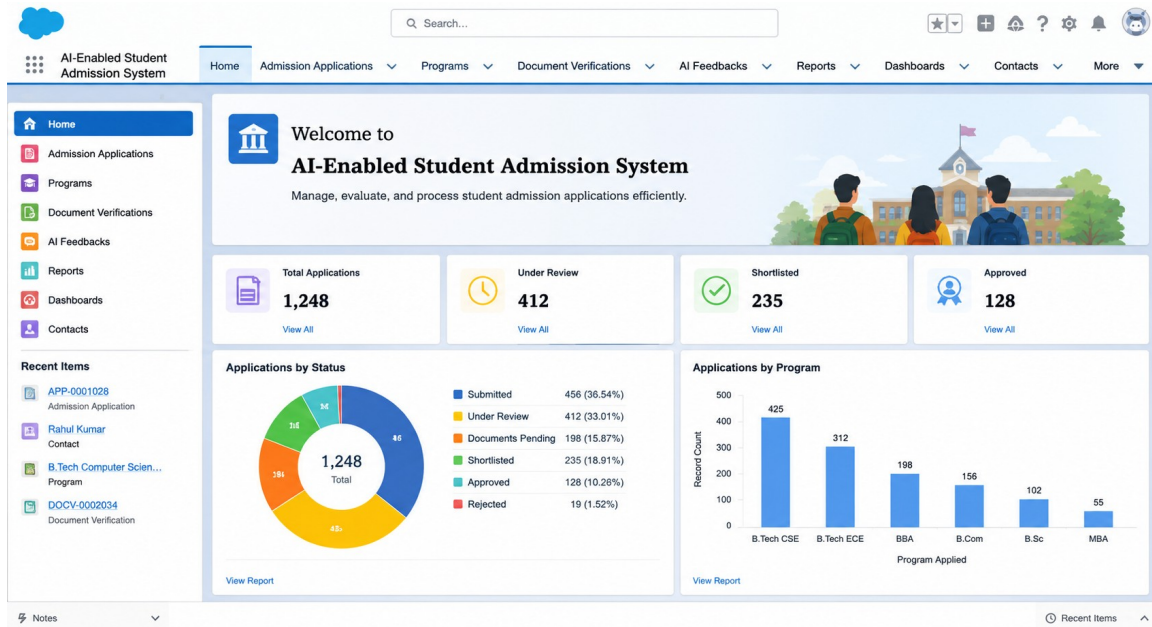


Figure 2: AI-Enabled Student Admission System Lightning App with admission management navigation.

Milestone 3 – Object Model

The data model combines Salesforce's standard objects with four purpose-built custom objects to capture everything the AI-enabled admission process needs.

Standard Objects Used

Object	Key Fields
Account	Account Name, Phone, Type (Educational Institution / Department information)
Contact	First Name, Last Name, Email, Phone (Applicant / Student contact information)

Custom Objects Created

Object	Key Fields
Admission_Application__c	Application Number, Applicant, Program Applied, Academic Percentage, Application Status, Application Priority, Application Completeness, Missing Documents, AI Admission Guidance, Application Date, Review Date, Admission Decision
Program__c	Program Name, Department, Eligibility Percentage, Available Seats, Admission Capacity, Active Status
Document_Verification__c	Verification Number, Admission Application, Document Type, Verification Status, Verification Comments, Verified By
AI_Feedback__c	AI Feedback Number (Auto Number), AI Suggestion Type, Feedback Status, Admission Officer Comments, Created By

Object Relationships

Relationship	Type
Contact → Admission Application	Lookup
Program → Admission Application	Lookup
Admission Application → Document Verification	Lookup
AI Feedback → Admission Application	Lookup

Custom tabs were created for each custom object (Program, Document Verification, and AI Feedback) so that admission officers and administrators could access and manage these records directly from the Lightning App navigation bar.

SETUP > OBJECT MANAGER

Program

Schema Builder Create

Details

Fields & Relationships
7 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Program Name	Program_Name__c	Text(100)	—	✓
Department	Department__c	Text(100)	—	—
Eligibility Percentage	Eligibility_Percentage__c	Number(5,2)	—	—
Available Seats	Available_Seats__c	Number(4,0)	—	—
Admission Capacity	Admission_Capacity__c	Number(4,0)	—	—
Active Status	Active_Status__c	Checkbox	—	✓
Created By	CreatedById	Lookup(User)	—	✓

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Restriction Rules
Triggers
Flow Triggers
Validation Rules

Figure 3: Program custom object used to manage academic program information and eligibility criteria.

SETUP > OBJECT MANAGER

Document Verification

Schema Builder Create

Details

Fields & Relationships
6 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Verification Number	Verification_Number__c	Text(50)	—	✓
Admission Application	Admission_Application__c	Lookup(Admission Application)	—	—
Document Type	Document_Type__c	Picklist	—	—
Verification Status	Verification_Status__c	Picklist	—	—
Verification Comments	Verification_Comments__c	Long Text Area(255)	—	—
Verified By	Verified_By__c	Lookup(User)	—	✓

Page Layouts
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Triggers
Flow Triggers
Validation Rules

Figure 4: Document Verification custom object for tracking applicant document verification.

Setup Home Object Manager

Search Setup

Schema Builder Create

SETUP > OBJECT MANAGER
AI Feedback

Details

Fields & Relationships
6 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED	
AI Feedback Number	AI_Feedback_Number__c	Text(50)	–	✓	▼
Admission Application	Admission_Application__c	Lookup(Admission Application)	–	–	▼
AI Suggestion Type	AI_Suggestion_Type__c	Picklist	–	–	▼
Feedback Status	Feedback_Status__c	Picklist	–	–	▼
Admission Officer Comments	Admission_Officer_Comments__c	Long Text Area(255)	–	–	▼
Created By	CreatedById	Lookup(User)	–	✓	▼

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Triggers

Flow Triggers

Validation Rules

Figure 5: AI Feedback custom object capturing admission officer feedback on AI-generated guidance.

Milestone 4 – Fields & Relationships (AI Feedback)

The AI_Feedback__c object was configured with the following fields to capture structured feedback on AI performance:

Field Label	API Name	Data Type	Purpose
AI Feedback Number	Name	Auto Number	Unique feedback record identifier
Admission Application	Admission_Application__c	Lookup (Admission Application)	Related admission application
Feedback Status	Feedback_Status__c	Picklist (Helpful, Not Helpful)	Officer rating of AI output
AI Suggestion Type	AI_Suggestion_Type__c	Picklist	Completeness Check / Eligibility Assessment / Admission Guidance
Admission Officer Comments	Admission_Officer_Comments__c	Long Text Area	Officer remarks on AI quality
Created By	CreatedById	Lookup (User)	Officer who provided feedback

Milestone 5 – Admission Application Record & Data Entry

A test Admission Application record was created to walk through the full lifecycle: entry of applicant details and program applied, automatic completeness check, eligibility evaluation, missing-document identification, and the AI Admission Guidance field, all visible on a single Lightning record page.

The screenshot displays the 'Admission Application' record for APP-0001028. The interface includes a top navigation bar with a search bar and various menu items. The main content area is divided into several sections:

- Header:** Shows the application ID (APP-0001028) and buttons for Follow, Edit, Clone, and Delete.
- Metadata:** Lists Applicant (Rahul Kumar), Program Applied (B.Tech Computer Science and Engineering), Application Status (Under Review), Application Date (May 12, 2025), and Application Officer (Annamneedi Satyanarayana).
- Progress Bar:** A horizontal bar showing the application's progress through stages: Submitted, Under Review (current), Documents Pending, Shortlisted, Approved, and Rejected. A 'Mark Status as Complete' button is at the end.
- Details Panel:**
 - Application Information:** Fields include Application Number (APP-0001028), Applicant (Rahul Kumar), Email (rahulkumar@email.com), Phone (+91 98765 43210), Application Date (May 12, 2025), Program Applied (B.Tech Computer Science and Engineering), and Department (Computer Science and Engineering).
 - Academic Information:** Fields include 10th Percentage (92.40%), 12th Percentage (88.20%), Academic Percentage (86.50%), Entrance Exam (JEE Main), and Entrance Exam Score (92.31 Percentile).
 - Application Status:** Fields include Application Status (Under Review), Application Priority (Medium), Application Completeness (Incomplete), Missing Documents (Transfer Certificate), Review Date, and Admission Decision.
 - AI & Guidance Information:** Includes AI Admission Guidance (with a note: 'Applicant meets the academic eligibility requirement. Transfer Certificate verification is pending. Request the applicant to upload the missing document before further processing.'), Last Guidance Generated On (May 12, 2025, 11:32 AM), Last Guidance Generated By (Agentforce AI Assistant), and AI Guidance Status (Generated).
- Right Sidebar:**
 - Document Verifications (2):** Lists 10th and 12th Marksheet, both with 'Verified' status.
 - AI Feedbacks (1):** Shows an AI Suggestion Type of 'Admission Guidance' with a 'Helpful' feedback status.
 - Notes (0):** A section for adding new notes.
 - Files (0):** A section for uploading files.

Figure 6: Admission Application record showing applicant information, application status, completeness, and AI Admission Guidance.

Example Record

- Application Number: APP-0001028
- Applicant: Rahul Kumar
- Program Applied: B.Tech Computer Science and Engineering
- Academic Percentage: 86.5%
- Application Status: Under Review
- Application Completeness: Incomplete
- Missing Documents: Transfer Certificate
- AI Admission Guidance: Applicant meets the academic eligibility requirement. Transfer Certificate verification is pending. Request the applicant to upload the missing document before further processing.

Milestone 6 – Validation Rules

A validation rule, `Academic_Percentage_Required`, was implemented on the Admission Application object to guarantee that no application can be submitted without the applicant's academic percentage — preventing incomplete evaluation caused by missing data.

Validation Formula

`AND( ISPICKVAL(Application_Status__c, "Submitted"), ISBLANK(Academic_Percentage__c) )`

- Error Message: “Academic Percentage must be provided before submitting the admission application.”
- Error Location: Academic Percentage field.
- Negative test: attempting to submit an application with a blank Academic Percentage correctly blocked the save.
- Positive test: entering a valid Academic Percentage allowed the application to save successfully.

The screenshot shows the Salesforce Setup interface with the 'Object Manager' tab selected. The 'Admission Application' object is chosen, and the 'Validation Rules' section is active. The rule 'Academic Percentage Required' is displayed with the following details:

- Rule Name:** Academic_Percentage_Required
- Active:** ☒
- Description:** Ensures academic percentage is provided before submitting an admission application.
- Error Condition Formula:** `AND( ISPICKVAL(Application_Status__c, "Submitted"), ISBLANK(Academic_Percentage__c) )`
- Error Message:** Academic Percentage must be provided before submitting the admission application.
- Error Location:** Top of Page
- Created By:** Annamneedi Satyanarayana, May 10, 2025, 10:15 AM
- Modified By:** Annamneedi Satyanarayana, May 10, 2025, 10:15 AM

The 'Validation Rule Formula' section shows the formula: `AND( ISPICKVAL(Application_Status__c, "Submitted"), ISBLANK(Academic_Percentage__c) )`.

The 'Validation Rule Test' table shows the results of the rule:

Test Case	Application Status	Academic Percentage	Expected Result	Actual Result
Negative Test Case	Submitted	— (Blank)	Error	✓ Error shown
Positive Test Case	Submitted	85.60	No Error	✓ Record saved

Figure 7: The Academic Percentage validation rule configured on the Admission Application object.

Milestone 7 – Automation: Record-Triggered Flow

A record-triggered flow was built to automatically check application completeness, evaluate program eligibility, and identify missing documents the moment an application is created or updated — removing this repetitive work from admission officers entirely.

- Object: Admission Application; Trigger: When a record is created or updated.
- Decision logic compares Academic Percentage against the selected Program's Eligibility Percentage.
- Decision logic checks related Document Verification records to determine whether required documents are present.
- Application Completeness is set to Complete or Incomplete, and missing documents are listed on the record.

- Application Status is updated (for example, to Documents Pending, Under Review, or Eligible) based on the evaluation outcome.
- The flow was activated and tested by creating applications across complete, incomplete, and ineligible scenarios, confirming correct status assignment in each case.

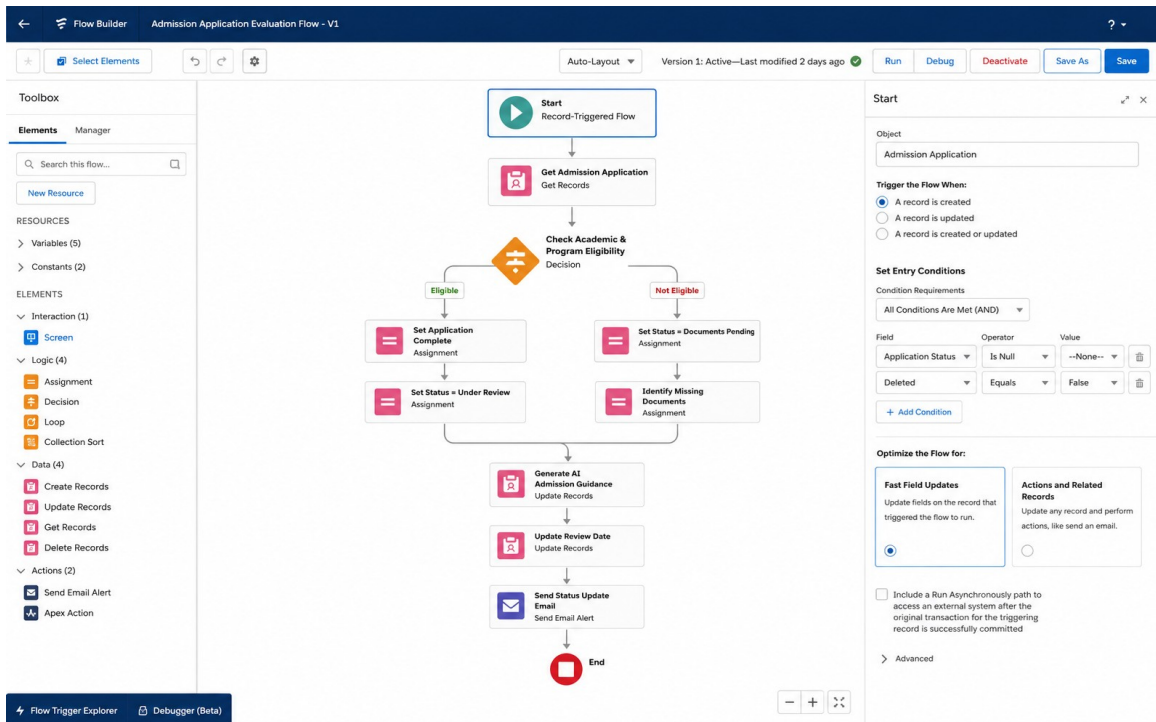


Figure 8: Record-triggered flow for admission application completeness and eligibility evaluation.

Milestone 8 – Email Templates & Alerts

To keep applicants informed as their admission status changes, an Admission Status Update email was implemented using a classic Email Template and an Email Alert, triggered by a record-triggered flow whenever an application's Application Status changes.

Email Template: Admission Status Update

- Subject: “Your Admission Application Status Has Been Updated – {! Admission_Application__c.Application_Number__c}”
- Body includes the applicant's name, application number, and current status, with a closing message directing the applicant to await further communication.

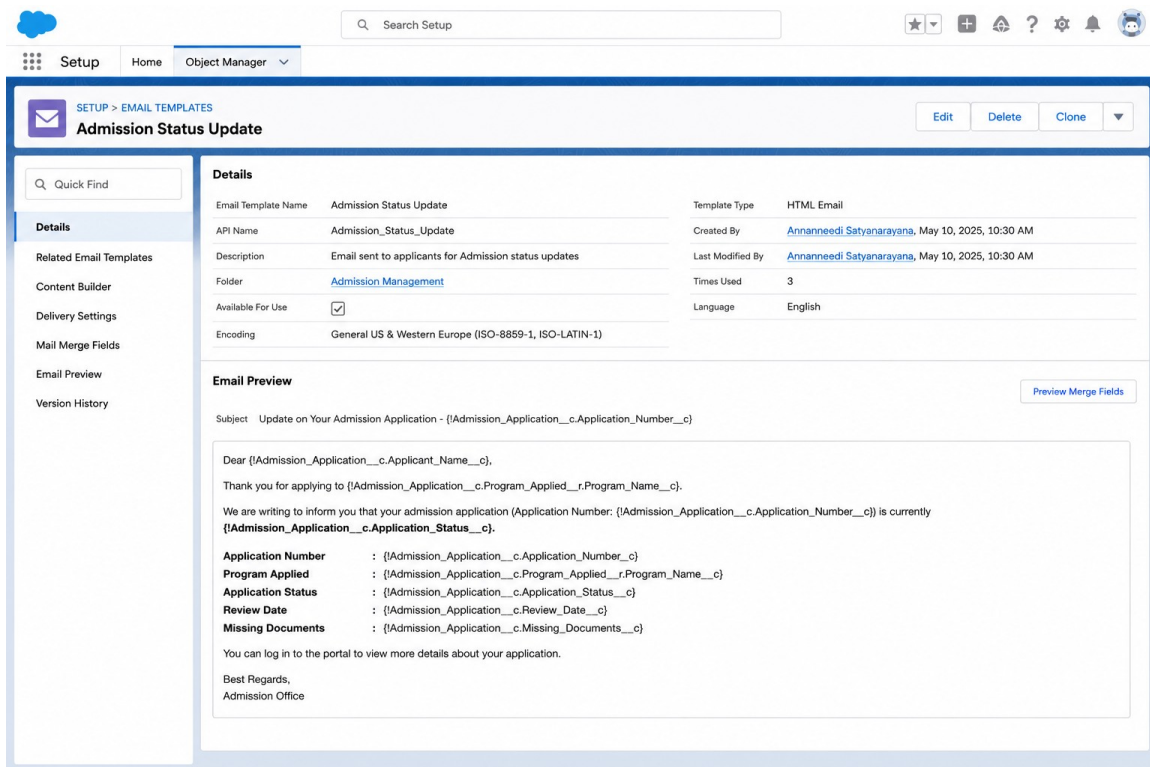


Figure 9: The Admission Status Update email template configured with applicant merge fields.

Email Alert Configuration

- Object: Admission Application; Email Template: Admission_Status_Update; Recipients: Contact Email.
- Triggered by a record-triggered flow with the condition Application Status Changed, using “Only when the record is updated to meet the condition requirements” to avoid duplicate emails.

Testing: An existing application with a status of Submitted was updated to Under Review. The flow fired correctly and the Admission Status Update email was delivered to the associated Contact's email address, confirming end-to-end automation.

7. Agentforce AI Implementation

The AI capability at the heart of this project is delivered through Agentforce, Salesforce's native AI agent platform, combined with Prompt Builder for structured, field-aware prompt generation. The AI Admission Assistant supports two functions: highlighting application completeness and eligibility, and generating admission guidance for the reviewing officer.

7.1 How Agentforce Works in This Project

- An Admission Application is created with Applicant, Program Applied, Academic Percentage, Application Status, and Missing Documents information.
- The Agentforce Agent, when triggered on the Admission Application record, reads the application details via a Prompt Template.
- The Prompt Template asks the model to act as an AI admission assistant and draft clear, professional guidance based on the applicant's academic percentage, program eligibility, completeness, and missing documents.
- The generated guidance is displayed to the officer, either directly on the Admission Application record (via the AI Admission Guidance field) or through the conversational Agentforce Agent panel.
- The admission officer reviews the guidance, and makes the final admission decision — the AI assistant never finalizes an admission decision on its own.

7.2 Enabling Agentforce and Prompt Builder

- Setup → Einstein Setup → toggled Turn on Einstein to enable AI features org-wide.
- Setup → Prompt Builder → enabled Prompt Templates.

7.3 Prompt Template: AI Admission Guidance

A Field Generation prompt template was created against the Admission_Application__c object, targeting a long-text field, AI_Admission_Guidance__c, added specifically to store the AI's generated guidance.

Prompt used:

“You are an AI admission assistant supporting a college admission officer. Review the following student admission application information: Program Applied: {!Admission_Application__c.Program_Applied__c}. Academic Percentage: {!Admission_Application__c.Academic_Percentage__c}. Application Status: {!Admission_Application__c.Application_Status__c}. Application Completeness: {!Admission_Application__c.Application_Completeness__c}. Missing Documents: {!Admission_Application__c.Missing_Documents__c}. Generate clear and professional admission guidance. Explain: 1. The current application status. 2. Whether the application information appears complete. 3. Any missing documents or information. 4. The recommended next action for the admission officer. 5. A professional suggested response that can be sent to the applicant. Do not make a final admission decision. The admission officer must review the recommendation.”

The template was saved, activated, and assigned as the org default, then verified directly on the Admission Application record by triggering the field's AI generation icon and confirming coherent, context-aware guidance was produced.

7.4 Agentforce Agent Configuration

The default Agentforce Agent was opened in Agent Builder and extended with a custom Topic and Action tailored to this use case:

Setting	Value
Topic Name	Student Admission Assistance
Classification Description	This topic handles student admission application guidance, applicant queries, eligibility information, missing documents, and admission status assistance.
Scope	Provide admission-related guidance based on student application information.

Agent Instructions

- Ask for the Admission Application ID when required.
- Retrieve relevant application information.
- Review applicant details and application status.
- Identify incomplete or missing information.
- Generate clear admission guidance.
- Provide a professional suggested response for the applicant.
- Never make a final admission decision without human admission officer review.

Agent Action

- Reference Action Type: Prompt Template; Reference Action: AI Admission Guidance.
- Agent Action Label: AI Admission Guidance.
- Instructions: first collect the Admission Application ID from the user, then behave as an AI admission assistant and return clear, helpful admission guidance.
- “Show in conversation” enabled so the generated guidance displays directly in the chat with the officer.

The agent was activated and tested with prompts such as “Can you give admission guidance for this application?”, which triggers the agent to request an Admission Application ID, then returns the AI-generated guidance for that specific application — confirmed working end-to-end inside the AI-Enabled Student Admission System app.

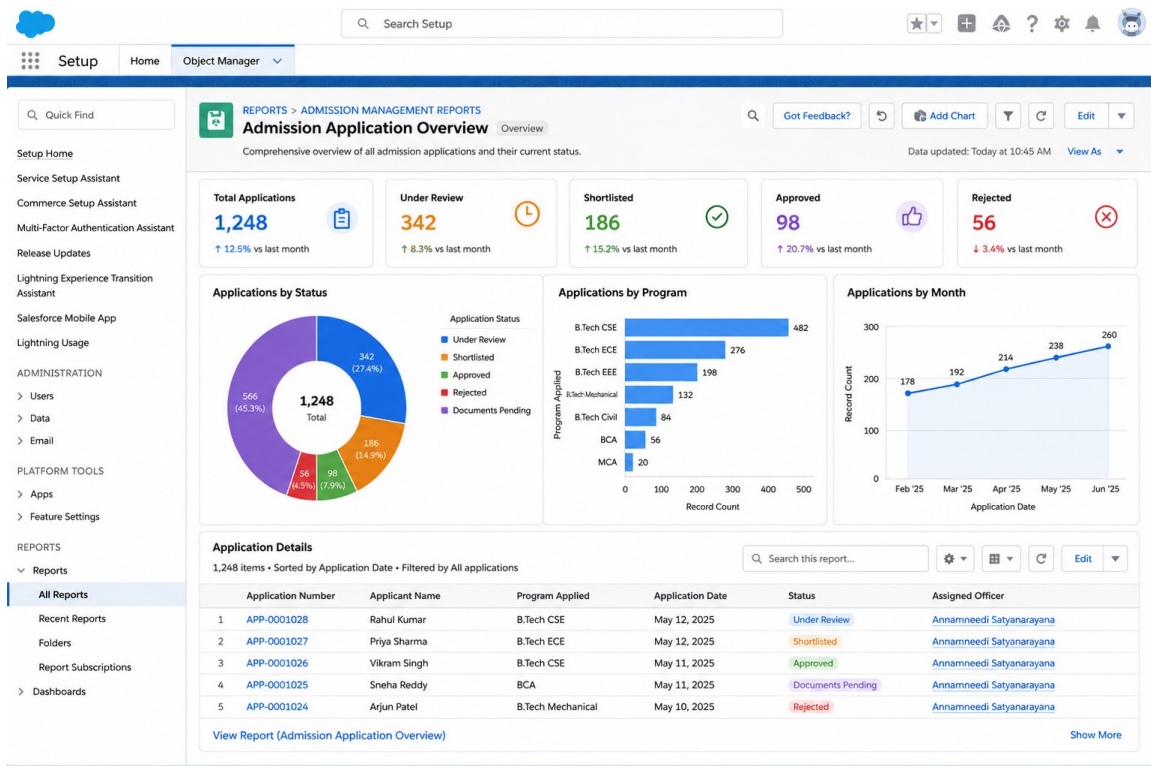


Figure 10: Agentforce AI Admission Assistant generating admission guidance.

7.5 Outcome

The combination of Prompt Builder and a custom Agentforce Topic/Action gives admission officers a conversational, on-demand way to retrieve AI-generated guidance for any application simply by supplying the Admission Application ID — without leaving the Salesforce interface. Officer feedback on this guidance is captured through the AI_Feedback__c object, closing the loop between AI output and measured usefulness.

8. Reports & Dashboards

Three reports were built to give admission managers real-time visibility into application volume, program demand, and officer review performance, all accessible from the AI-Enabled Student Admission System app.

8.1 Applications by Status

- Report Type: Admission Applications.
- Columns: Application Number, Applicant, Program Applied, Application Status, Application Date.
- Grouped by Application Status (Submitted, Under Review, Documents Pending, Shortlisted, Approved, Rejected) to give an immediate view of how applications are distributed across the pipeline.

8.2 Applications by Program

- Report Type: Admission Applications.
- Columns: Application Number, Program Applied, Applicant, Application Status.
- Grouped by Program Applied to show the number of applications received for each academic program.

8.3 Application Review Performance

- Report Type: Admission Applications; Filter: Review Date is not blank.
- Columns: Application Number, Reviewed By, Application Status, Application Date, Review Date.
- Grouped by Reviewed By (admission officer), to reveal how many applications each officer has reviewed and how many remain pending.

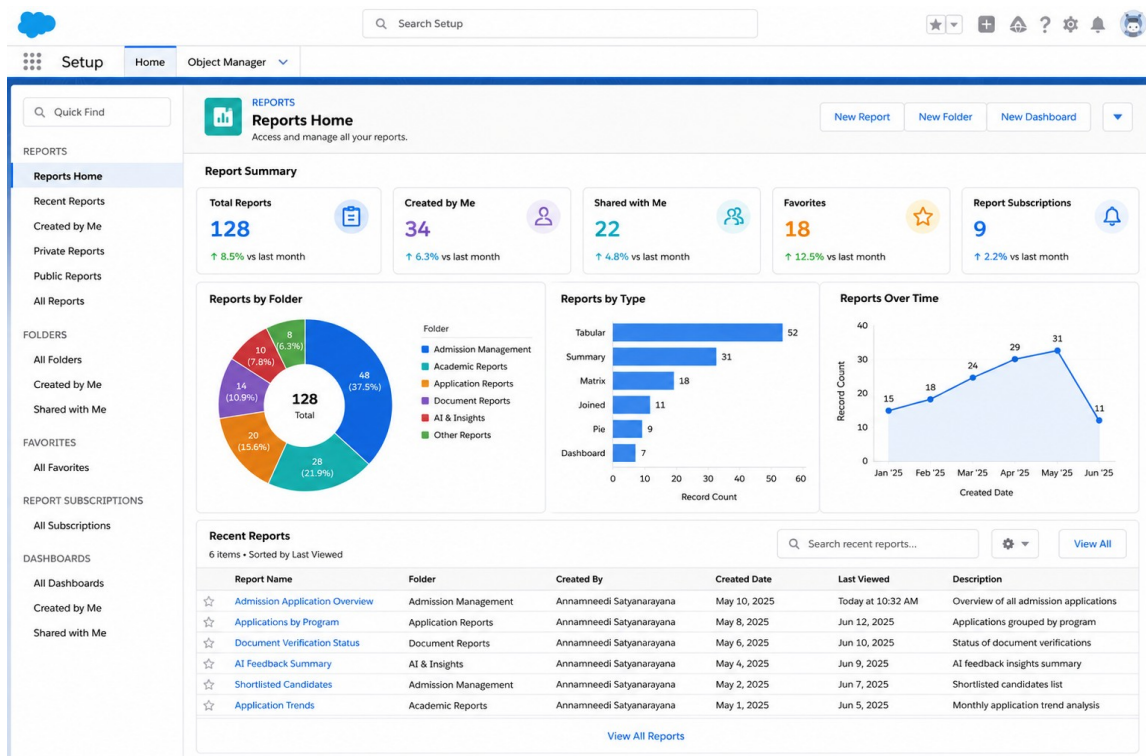


Figure 11: Admission reports available in the Admission Reports folder.

8.4 Admission Management Dashboard

An “Admission Management Dashboard” was created to give admission managers a single-glance view of application distribution and review performance, refreshed in real time as application data changes.

- Applications by Status – Donut Chart.
- Applications by Program – Bar Chart.
- Admission Review Performance – Summary Chart.

These dashboard components give admission managers real-time visibility into how applications are progressing through the pipeline, which programs are attracting the most interest, and how review work is distributed across the admission team, supporting timely, data-informed decisions.

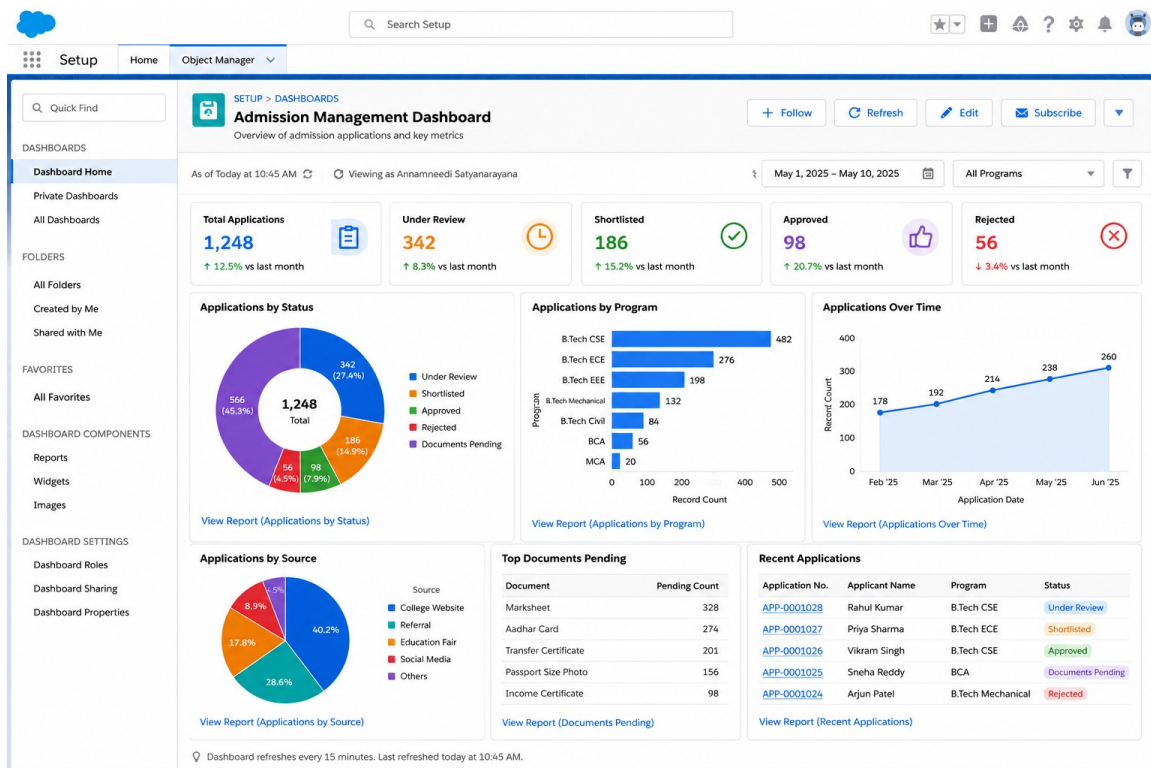


Figure 12: Admission Management Dashboard displaying application status and review analytics.

9. Security & Access Control

Role-based access control was implemented using cloned profiles for the two primary user roles — Admission Officer and Admission Manager — alongside dedicated permission sets that grant additional, feature-specific access. The System Administrator profile was used as-is for full administrative access.

9.1 Admission Officer Profile

Cloned from Standard User. Configured with:

- Admission Application: Read, Create, and Edit enabled; Delete disabled.
- Program: Read access.
- Document Verification: Read, Create, and Edit enabled.
- AI_Feedback__c: Create and Read enabled; Edit and Delete disabled.
- Field-level security: Application Status and Missing Documents editable; AI Admission Guidance read-only.
- App access: AI-Enabled Student Admission System enabled.

Profile: Admission Officer

Users with this profile have the permissions and page layouts listed below. Administrators can change a user's profile by editing the user record.

Profile Detail

Name	Admission Officer	Custom Profile	✓
User License	Salesforce	Modified By	Admin User, 5/1/2025, 2:30 PM
Description	Profile for Admission Officers to manage admission applications and related records.	Profile Used	5
Created By	Admin User, 4/25/2025, 10:15 AM	API Name	Admission_Officer

Object Settings

Object	Tab Settings	Application	Read	Create	Edit	Delete	View All	Modify All
Admission Application	Default On	AI-Enabled Student Admission System	✓	✓	✓	—	✓	—
Program	Default On	AI-Enabled Student Admission System	✓	—	—	—	✓	—
Document Verification	Default On	AI-Enabled Student Admission System	✓	✓	✓	—	✓	—
AI Feedback	Default On	AI-Enabled Student Admission System	✓	✓	—	—	✓	—
Contact	Default On	AI-Enabled Student Admission System	✓	—	—	—	✓	—

Field-Level Security

Field	Admission Application	Program	Document Verification	AI Feedback
Application Status	Read-Write	—	—	—
Academic Percentage	Read-Write	—	—	—
AI Admission Guidance	Read-Only	—	—	—
Missing Documents	Read-Write	—	—	—
Admission Decision	Read-Write	—	—	—

Figure 13: Admission Officer profile configured with admission-specific object permissions.

9.2 Admission Manager Profile

Cloned from Standard User. Configured with:

- Admission Application: Read and Edit enabled.
- Program: Read and Edit enabled.
- Document Verification: Read access.
- AI_Feedback__c: Read access.
- Reports & Dashboards: Create and Subscribe access enabled.

- Field-level security: Admission Decision editable; AI Admission Guidance read-only.
- App access: AI-Enabled Student Admission System enabled.

Admission Manager Profile

Profile Overview

Description: Profile for Admission Managers to monitor and manage admission applications and reports.

User License: Salesforce

Custom Profile: ☒

Profile Name: Admission Manager Profile

Profile API Name: Admission_Manager_Profile

Created By: Annamneedi Satyanarayana, 5/15/2025, 10:20 AM

Last Modified By: Annamneedi Satyanarayana, 5/15/2025, 02:18 PM

Object Permissions

Object	Read	Create	Edit	Delete	View All / Modify All
Admission Application	☒	☒	☒		☒
Program	☒	☒	☒		☒
Document Verification	☒				☒
AI Feedback	☒				
Account	☒				
Contact	☒				
Reports	☒	☒	☒		☒
Dashboard	☒	☒	☒		☒

Field-Level Security

Field	Read Access	Edit Access
AI Admission Guidance (Admission_Application__c)	☒	(Controlled by FLS)
Academic Percentage (Admission_Application__c)	☒	☒
Application Status (Admission_Application__c)	☒	☒
Missing Documents (Admission_Application__c)	☒	☒
Admission Decision (Admission_Application__c)	☒	☒

Permissions Summary

- Login Access: ☒
- API Enabled: ☒
- Modify All Data: ☒
- View All Data: ☒
- Author Apex: ☒
- Manage Users: ☒

Assigned Users

2
[View Users](#)

Figure 14: Admission Manager profile configured for application monitoring and reporting.

9.3 Permission Sets

Two permission sets extend the base profiles with feature-specific access, following the recommended Salesforce pattern of using profiles for baseline access and permission sets for additional capabilities.

Permission Set	Grants	Assigned To
AI Admission Feature Access	Edit AI-related Admission Application fields; Create/Read on AI_Feedback__c; Access Einstein Features; Use Agentforce	Admission Officers
Admission Report Access	Create and Customize Reports; Subscribe to Reports; Dashboard Access	Admission Managers

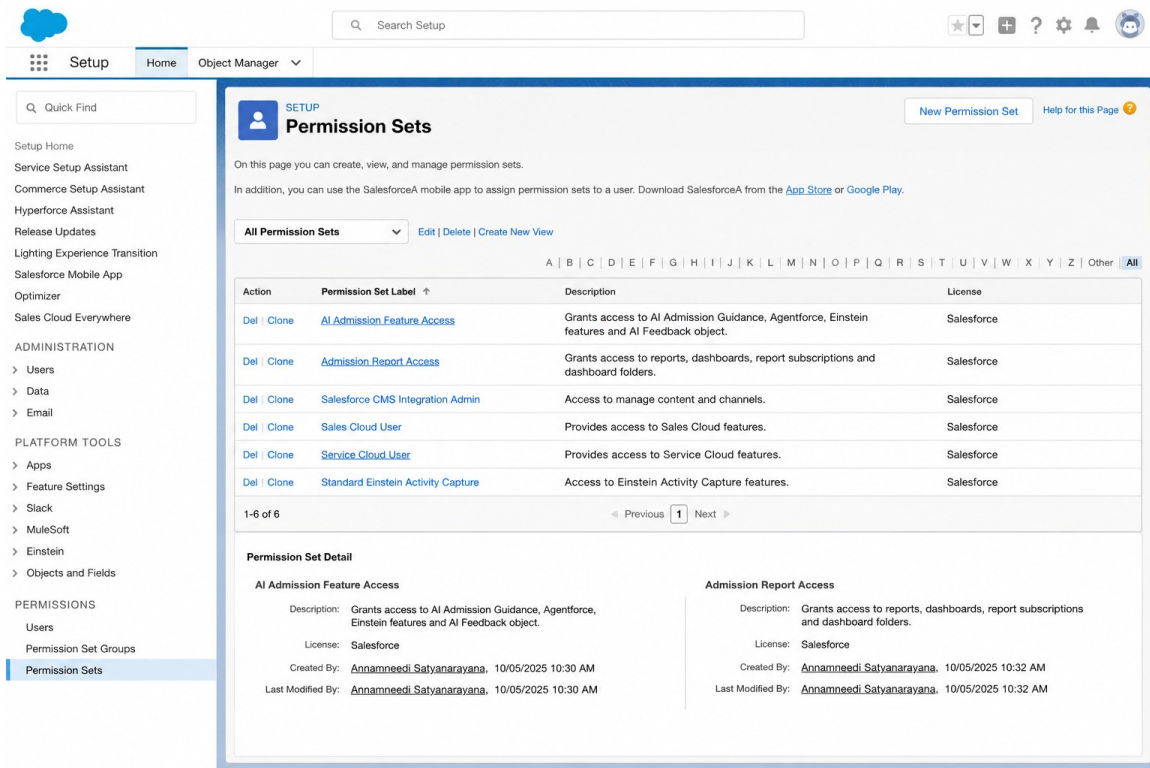


Figure 15: AI Admission Feature Access and Admission Report Access permission sets.

9.4 Role – Profile – Permission Set Summary

Role	Profile	Permission Set
Admission Officer	Admission Officer	AI Admission Feature Access
Admission Manager	Admission Manager	Admission Report Access
Admin	System Administrator	All (native)

9.5 Users

Three users were created to represent the distinct roles in the system, each assigned the appropriate profile and permission set:

User	Profile	Permission Set Assigned
Admin User	System Administrator	—
Admission Officer	Admission Officer	AI Admission Feature Access
Admission Manager	Admission Manager	Admission Report Access

Final verification confirmed that each user has exactly one profile, is marked Active, has a unique username, and that the assigned profile correctly matches the intended role — satisfying the security design of the system.

10. Functional & Performance Testing

Testing was performed throughout the development phase to verify that each automation, validation rule, and AI feature behaved as designed. The following scenarios were executed and verified directly in the Salesforce Developer Org:

- Application creation across complete, incomplete, and ineligible scenarios, confirming correct completeness and eligibility evaluation.
- Validation rule testing: attempting to submit an application with a blank Academic Percentage was correctly blocked; supplying a value allowed the save to succeed.
- Agentforce testing: triggering the AI Admission Guidance action with a valid Admission Application ID returned coherent, context-appropriate guidance drafted from the application's program, academic percentage, and completeness.
- AI Feedback testing: logging Helpful / Not Helpful feedback against an application correctly created an AI_Feedback__c record linked to that application.
- Email alert testing: updating an application's Application Status triggered the Admission Status Update flow, and the email was delivered to the Contact's email address.
- Reports and dashboard testing: confirmed that Applications by Status, Applications by Program, and Application Review Performance reports reflected live application data accurately, and that the dashboard refreshed correctly.
- Security testing: logged in as each of the three test users to confirm object, field, and app-level access matched the intended profile and permission set configuration.

Performance testing confirmed that record-triggered flows executed instantly on save, and that reports and dashboards rendered within a few seconds even as application volume increased, with no noticeable delay for end users.

11. Maintenance, Monitoring & Troubleshooting

The Salesforce AI-Enabled Student Admission System requires ongoing maintenance and monitoring to ensure smooth operation, data accuracy, and reliable automation. Regular maintenance activities include reviewing and updating custom objects, fields, picklist values, and flows to align with changing institutional requirements. Profiles and permission sets are periodically audited to ensure proper access control and compliance with security best practices.

System monitoring focuses on tracking application volumes, program-wise demand, and automation performance using reports and dashboards. Flow executions are reviewed to confirm that completeness checks, eligibility evaluations, and document verification are functioning as expected. AI feedback data is monitored to evaluate the effectiveness of system-generated guidance and identify areas for improvement.

Troubleshooting involves analyzing flow errors, validation failures, and data inconsistencies when issues occur. Administrators review debug logs, flow error emails, and record history to identify root causes. Corrective actions may include updating flow logic, adjusting field-level security, refining decision conditions, or correcting data configurations. This structured approach ensures system stability, minimizes downtime, and supports continuous improvement of the admission process.

12. Advantages & Disadvantages

12.1 Advantages

- Centralized management of all student admission applications in a single Salesforce app.
- Reduced manual effort through automated completeness verification, eligibility checks, and missing-document identification.
- Faster, more informed officer reviews through AI-generated admission guidance via Agentforce.
- Improved visibility for admission managers into application volume and program demand through real-time reports and dashboards.
- Strong data protection through role-based profiles, permission sets, and field-level security.
- A built-in feedback loop that measures and can improve AI guidance quality over time.
- Built entirely on native Salesforce Admin and Flow features, keeping the solution low-cost and easy to maintain.

12.2 Disadvantages

- Requires an active Salesforce license and Agentforce/Einstein feature enablement, which may carry cost implications in a production org.
- Fully dependent on internet connectivity as a cloud-based platform.
- AI-generated guidance still requires officer review before an admission decision is made, so full automation of decisions is intentionally not supported.
- Limited offline access, consistent with other cloud-native Salesforce applications.

13. Conclusion

The Salesforce AI-Enabled Student Admission System successfully demonstrates how intelligent automation and structured data modeling can improve the efficiency of student admission processing. By leveraging standard and custom Salesforce objects, record-triggered flows for completeness verification, eligibility evaluation, and document tracking, and a well-defined security model, the system reduces manual effort and supports timely, consistent application processing.

The inclusion of Agentforce-powered admission guidance and a structured feedback mechanism enhances officer productivity while enabling continuous improvement of the system's AI guidance. Admission officers remain responsible for reviewing applications and making the final admission decision at every stage. Overall, the project delivers a scalable, maintainable admission management solution, and provides strong hands-on experience with core Salesforce Admin, Flow automation, and Agentforce concepts — making it well suited for both practical implementation and interview demonstration.

14. Future Scope

- Extend Agentforce with multi-turn conversational application guidance, allowing applicants to ask about their status directly before an officer takes over.
- Introduce a knowledge base integration so AI-generated guidance can reference institutional admission policies and FAQs.

- Add escalation automation that flags applications automatically when a review is pending beyond an expected timeframe.
- Build a mobile-optimized officer experience for on-the-go application review.
- Expand the AI feedback loop into a lightweight analytics dashboard tracking guidance accuracy trends over time.
- Integrate additional application-intake channels (web portal, email, and mobile) to broaden how applicants can apply.

15. Appendix

Item	Details
Source	Salesforce Configuration — Objects, Flows, Validation Rules, Agentforce, Prompt Builder
Dataset	Sample Admission Application, Contact, Program, and Account records created during testing
Tools Used	Salesforce Developer Org, Flow Builder, Agentforce Agent Builder, Prompt Builder, Reports & Dashboards
Team	Annamneedi Satyanarayana (Individual Project)

Thank You